

Review of Literature

Validation of Accessible Posturography Tools

Although they provide highly accurate posturographic data, traditional laboratory-grade force plates are quite expensive, bulky, and usually confined to clinical or dedicated research settings. To address this barrier, researchers have begun to turn to more affordable and commercially available alternatives such as the video gaming device, the Nintendo Wii Balance Board (WBB). When its questionable practicality was tested in a comprehensive systematic review, Clark et al. (2018) demonstrated that the WBB is a highly reliable and valid instrument for assessing standing balance, even when compared to gold-standard laboratory equipment. This validation has enabled the WBB's use in various other functional and observational studies. An example of such would be Tapanya and Sangkarit (2024) and their success in utilizing the WBB to investigate the detrimental effects of smartphone usage on postural stability, observing significantly increased sway in individuals with pre-existing forward head posture.

Machine Learning Applications in Postural Analysis

Beyond simply collecting center-of-pressure and balance data, the machine learning (ML) algorithms are also able to interpret complex postural metrics. Seo et al. (2023) developed a stability index estimation algorithm that applies ML to the data derived from the WBB and a custom pressure mat. Their models were able to accurately diagnose overall balance stability in elderly populations, demonstrating the capability of affordable hardware and algorithmic approaches to evaluate everyday balance in non-clinical environments. Similarly, Kang, Hwang, and Etebar Zadeh (2026) expanded the use of ML in occupational ergonomics by predicting both objective and subjective lower back fatigue during sustained trunk flexion. Their findings

underscore the potential of ML models to continuously track muscular fatigue without relying on intrusive, traditional medical sensors.

Clinical Diagnostics and Disease Tracking

The combination of accessible hardware and advanced data analysis has also proven highly effective in specialized clinical diagnostics. Severini et al. (2017) investigated the WBB's diagnostic sensitivity in patients with Multiple Sclerosis (MS) who exhibited only minimal balance impairments. Despite slight baseline discrepancies in raw values compared to laboratory-grade force plates, the WBB maintained over 80% accuracy in distinguishing between healthy individuals and MS patients. This firmly established the device's utility as a low-cost tool for classifying distinct populations and tracking disease progression over time.

Research Gap and Current Study

Collectively, current literature has established that the WBB is a valid, low-cost tool for measuring balance (Clark et al., 2018; Tapanya & Sangkarit, 2024) and that machine learning is effective in interpreting this data to classify postures and clinical balance impairments (Seo et al., 2023; Kang et al., 2026; Severini et al., 2017). However, there is still a gap in applying these technologies directly to everyday occupational ergonomics. While standing workstations are being increasingly adopted to combat the adverse musculoskeletal effects of prolonged sitting and bad posture, there is a lack of accessible, real-time monitoring systems designed to classify working postures and detect slouching as users fatigue throughout the workday. The present study addresses this gap by using WBB sensor data and ML classification models (Logistic Regression, Decision Tree, and Random Forest) to identify and classify standing postures at workstations, thereby creating a cost-effective solution for continuous ergonomic monitoring and intervention.